

DS	Session	Arrays	Question Information	Level 1 Challenge 21
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Problem Description:

How many Y's did a Roman Centurion make a day in cold hard Lira? About a C's worth! Turns out, Martians gave Rome the idea for their number system. Use the conversion charts below to help translate some Martian numbers!

Note, that unlike the Roman Numerals, Martian Numerals reuse symbols to mean different values. B can either mean '1' or '100' depending on where it appears in the number sequence.

Input Format:

You will receive a list of numbers in a data file, one number per line, up to 5 lines at a time (with a minimum of 1 line). No number will exceed 1000, or be less than 1.

Output Format:

Print the output in a separate lines contains convert the numbers from Arabic (1,2,3...10...500...1000) to Martian (B,BB,BBB...Z...G...R) numerals.

▼ Logical Test Cases

Test Case 1	Test Case 2
INPUT (STDIN)	INPUT (STDIN)
26	161
67	627

186
408
500

EXPECTED OUTPUT

ZZWB
PZWBB
BPZZZWB
BGWBBB
G

816
401
501

EXPECTED OUTPUT

BPZB
GBZZWBB
GBBBZWB
BGB
GB

✓ Mandatory Test Cases

Test Case 1

KEYWORD

`char buf[]`

Test Case 2

KEYWORD

`buf[i++]='R';`

Test Case 3

KEYWORD

`while(n>=10)`

✓ Complexity Test Cases

Test Case 1

CYCLOMATIC COMPLEXITY

18

Test Case 2

TOKEN COUNT

455

Test Case 3

NLOC

84

```
1 #include <stdio.h>
2 #include <string.h>
3
4 void martian_conv(int num) {
5     char buf[100];
6     int i = 0;
7     int values[] = {1000, 900, 500, 400, 100, 90, 50, 40, 10, 9, 5, 4, 1};
8     char *symbols[] = {"R", "BR", "G", "BG", "B", "ZB", "P", "ZP", "Z", "BZ", "W", "BW", "B"};
9
10    for (int j = 0; j < 13; j++) {
11        while (num >= values[j]) {
12            int len = strlen(symbols[j]);
13            for (int k = 0; k < len; k++) {
14                buf[i++] = symbols[j][k];
15            }
16            num -= values[j];
17
18        }
19        buf[i] = '\0';
20        printf("%s\n", buf);
21    }
22
23    int main() {
24        int n;
25        int count = 0;
```

```
26  while (scanf("%d", &n) != EOF && count < 5) {  
27      martian_conv(n);  
28      count++;  
29  }
```

```
30     return 0;  
31}  
32
```

C:\Users\CH.SC.U4CSE25225\Desktop\DSA Lab>type t1.txt

26

67

186

408

500

C:\Users\CH.SC.U4CSE25225\Desktop\DSA Lab>Array1.exe

ZZWB

PZWBB

BPZZWB

BGWBBB

G

C:\Users\CH.SC.U4CSE25225\Desktop\DSA Lab>

C:\Users\CH.SC.U4CSE25225\Desktop\DSA Lab>type t2.txt

161

627

816

401

501

C:\Users\CH.SC.U4CSE25225\Desktop\DSA Lab>Array1.exe

BPZB

GBZZWBB

GBBBZWB

BGB

GB

C:\Users\CH.SC.U4CSE25225\Desktop\DSA Lab>

Problem Description: Ram has provide inputs two numbers 'p' and 'q' to Sakthi. He wants to creates a matrix of size $p \times q$ (p rows and q columns) in which every elements is either Y or O. The Ys and Os must be filled alternatively, the matrix should have outermost rectangle of Ys, then a rectangle of Os, then a rectangle of Ys, and so on..

Constraints:

$1 \leq p, q \leq 1000$

Input Format:

Input lines must be how many rows and columns in that matrix, also values must be separate space.

Output Format:

Print the output in a separate lines.

▼ Logical Test Cases

Test Case 1

INPUT (STDIN)

6 7

Test Case 2

INPUT (STDIN)

5 8

EXPECTED OUTPUT

```
Y Y Y Y Y Y Y
Y 0 0 0 0 0 Y
Y 0 Y Y Y 0 Y
Y 0 Y Y Y 0 Y
Y 0 0 0 0 0 Y
Y Y Y Y Y Y Y
```

EXPECTED OUTPUT

```
Y Y Y Y Y Y Y Y
Y 0 0 0 0 0 0 Y
Y 0 Y Y Y Y 0 Y
Y 0 0 0 0 0 0 Y
Y Y Y Y Y Y Y Y
```

✓ Mandatory Test Cases

Test Case 1

KEYWORD

```
while(top<=bottom &&
right>=left)
```

✓ Complexity Test Cases

Test Case 1

CYCLOMATIC COMPLEXITY

12

Test Case 2

TOKEN COUNT

290

Test Case 3

NLOC

52

```
1 #include <stdio.h>
2
3 int main() {
4     int rows, cols;
5     if (scanf("%d %d", &rows, &cols) != 2) return 0;
6     char arr[rows][cols];
7     int left = 0, right = cols - 1, top = 0, bottom = rows - 1;
8     int count = 0;
9     while (top <= bottom && right >= left) {
10         char c = (count % 2 == 0) ? 'Y' : '0';
11         for (int i = left; i <= right; i++) {
12             arr[top][i] = c;
13         }
14         top++;
15         for (int i = top; i <= bottom; i++) {
16             arr[i][right] = c;
17         }
18         right--;
19         if (top <= bottom) {
20             for (int i = right; i >= left; i--) {
21                 arr[bottom][i] = c;
22             }
23             bottom--;
24         }
25         if (left <= right) {
```

```
26         for (int i = bottom; i >= top; i--) {
27             arr[i][left] = c;
28         }
29         left++;
30     }
31     count++;
32 }
33 for (int i = 0; i < rows; i++) {
```

```
34     for (int j = 0; j < cols; j++) {
35         printf("%c ", arr[i][j]);
36     }
37     printf("\n");
38 }
39 return 0;
40}
41
```

C:\Users\CH.SC.U4CSE25225\Desktop\DSA Lab>type t1.txt

6 7

C:\Users\CH.SC.U4CSE25225\Desktop\DSA Lab>Array2.exe

Y Y Y Y Y Y

Y 0 0 0 0 0 Y

Y 0 Y Y Y 0 Y

Y 0 Y Y Y 0 Y

Y 0 0 0 0 0 Y

Y Y Y Y Y Y Y

C:\Users\CH.SC.U4CSE25225\Desktop\DSA Lab>

C:\Users\CH.SC.U4CSE25225\Desktop\DSA Lab>type t2.txt

5 8

C:\Users\CH.SC.U4CSE25225\Desktop\DSA Lab>Array2.exe

Y Y Y Y Y Y Y

Y 0 0 0 0 0 0 Y

Y 0 Y Y Y Y 0 Y

Y 0 0 0 0 0 0 Y

Y Y Y Y Y Y Y Y

C:\Users\CH.SC.U4CSE25225\Desktop\DSA Lab>

DS	Session	Arrays	Question Information	Level 1	Challenge 23
Problem Description: Rigesh is an electronic shop owner. Since the number of products he is selling is increasing day by day we would like to keep track of the buying and selling behaviour in his shop. So given the cost of stock on each day in an array $A[]$ of size N. Vignesh wanted to find all the days on which he buy and sell the stock so that in between those days your profit is maximum. Constraints: $1 \leq t \leq 10$ $1 \leq n \leq 10$ Input Format: First line contains number of test cases T. First line of each test case contains an integer value N denoting the number of days, followed by an array of stock prices of N days. Output Format: For each testcase, output all the days with profit in a single line. If there is no profit then print "No Profit".					

Logical Test Cases

Test Case 1

INPUT (STDIN)

1
5
130 204 150 175 140

EXPECTED OUTPUT

(0 1)(2 3)

Test Case 2

INPUT (STDIN)

2
5
131 124 110 175 110
3
23 45 67

EXPECTED OUTPUT

(2 3)
(0 2)

Mandatory Test Cases

Test Case 1

KEYWORD

if(arr[i]>arr[i-1])

Complexity Test Cases

Test Case 1

CYCLOMATIC COMPLEXITY

17

Test Case 2

TOKEN COUNT

350

Test Case 3

NLOC

71

```
1 #include <stdio.h>
2
3 int main() {
4     int T;
5     if (scanf("%d", &T) != 1) return 0;
6     while (T--) {
7         int n;
8         if (scanf("%d", &n) != 1) break;
9         int arr[n];
10        for (int i = 0; i < n; i++) {
11            scanf("%d", &arr[i]);
12        }
13        int i = 0, count = 0;
14        while (i < n - 1) {
15            while ((i < n - 1) && (arr[i + 1] <= arr[i]))
16                i++;
17            if (i == n - 1)
```

```
19            int buy = i++;
20            while ((i < n) && (arr[i] >= arr[i - 1])) {
21                if (arr[i] > arr[i - 1]) {}
22                i++;
23            }
24            int sell = i - 1;
25            printf("(%d %d)", buy, sell);
```

```
26         count++;
27     }
28     if (count == 0)
29         printf("No Profit");
30     printf("\n");
```

```
31  }  
32  return 0;  
33}  
34
```

```
C:\Users\CH.SC.U4CSE25225\Desktop\DSA Lab>type t1.txt
```

```
1
```

```
5
```

```
130 204 150 175 140
```

```
C:\Users\CH.SC.U4CSE25225\Desktop\DSA Lab>Array3.exe
```

```
(0 1)(2 3)
```

```
C:\Users\CH.SC.U4CSE25225\Desktop\DSA Lab>
```

```
C:\Users\CH.SC.U4CSE25225\Desktop\DSA Lab>type t2.txt
```

```
2
```

```
5
```

```
131 124 110 175 110
```

```
3
```

```
23 45 67
```

```
C:\Users\CH.SC.U4CSE25225\Desktop\DSA Lab>Array3.exe
```

```
(2 3)
```

```
(0 2)
```

```
C:\Users\CH.SC.U4CSE25225\Desktop\DSA Lab>
```

TCS Interview Question

Question description

Ravi participated in TCS off campus interview at reputed institution, one of the technical question he has to complete with in the given time, where you need to sort the array in the waveform. There might be multiple possible output of the program. the following pattern output is appreciated.

Function Description

This is a simple method of solving this question which contains basic 2 steps and they are as follow

Step : 1 – Sort the array in ascending order.

Step : 2 – Swap all adjacent elements of the array

Let us consider the input array be {3, 6, 5, 10, 7, 20}. After sorting, we get {3, 5, 6, 7, 10, 20}. After swapping adjacent elements, we get {5, 3, 7, 6, 20, 10}.

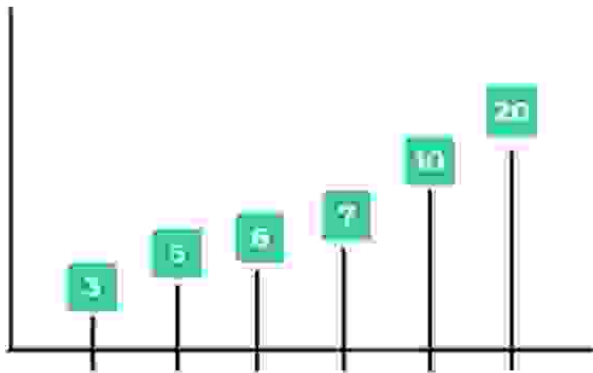
Brute Force Method : Step 1

Array

3 6 5 10 7 20

Step : 1

3 5 6 7 10 20



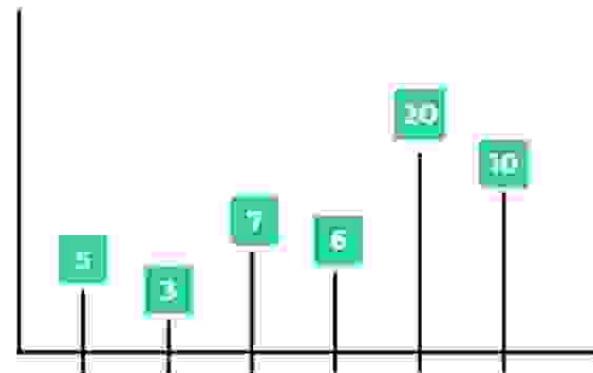
Brute Force Method : Step 2

Step : 1

3 5 6 7 10 20

Step : 2

5 3 7 6 20 10



Constraints

$0 < n < 100$

$0 < arr[i] < 1000$

Input Format

First line contains the value of n that is the total number of elements in the array.

-Second line contains the space separated elements of array.

Output Format

-Output contains only line that is only the resultant array in the wave form fashion.

Logical Test Cases

Test Case 1

INPUT (STDIN)

6
3 6 5 10 7 20

EXPECTED OUTPUT

5 3 7 6 20 10

Test Case 2

INPUT (STDIN)

8
10 5 6 3 2 20 98 80

EXPECTED OUTPUT

3 2 6 5 20 10 98 80

Mandatory Test Cases

Test Case 1

KEYWORD

int array[n]

Test Case 2

KEYWORD

for(int i=0;i<n;i++)

Test Case 3

KEYWORD

if(array[i]>array[j])

Complexity Test Cases

Test Case 1

CYCLOMATIC COMPLEXITY

7

Test Case 2

TOKEN COUNT

225

Test Case 3

NLOC

38

```
1 #include <stdio.h>
2
3 int main() {
4     int n;
5     if (scanf("%d", &n) != 1) return 0;
6     int array[n];
7     for (int i = 0; i < n; i++) {
8         scanf("%d", &array[i]);
9     }
10    for (int i = 0; i < n; i++) {
11        for (int j = i + 1; j < n; j++) {
12            if (array[i] > array[j]) {
13                int temp = array[i];
14                array[i] = array[j];
15                array[j] = temp;
16            }
17        }
18    }
19    for (int i = 0; i < n - 1; i += 2) {
20        int temp = array[i];
21        array[i] = array[i + 1];
22        array[i + 1] = temp;
23    }
24    for (int i = 0; i < n; i++) {
25        printf("%d%c", array[i], (i == n - 1) ? '\n' : ' ');
```

```
26     }  
27     return 0;
```

28}

29

```
C:\Users\CH.SC.U4CSE25225\Desktop\DSA Lab>type t1.txt
```

```
6
```

```
3 6 5 10 7 20
```

```
C:\Users\CH.SC.U4CSE25225\Desktop\DSA Lab>Array4.exe
```

```
5 3 7 6 20 10
```

```
C:\Users\CH.SC.U4CSE25225\Desktop\DSA Lab>
```

```
C:\Users\CH.SC.U4CSE25225\Desktop\DSA Lab>type t2.txt
```

```
8
```

```
10 5 6 3 2 20 98 80
```

```
C:\Users\CH.SC.U4CSE25225\Desktop\DSA Lab>Array4.exe
```

```
3 2 6 5 20 10 98 80
```

```
C:\Users\CH.SC.U4CSE25225\Desktop\DSA Lab>
```

DS	Session	Arrays	Question Information	Level 1	Challenge 25
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Problem Description:

For some reason, your school's football team has chosen to spell out the numbers on their jerseys instead of using the usual digits. Being great fans, you're going to be ready to cheer for your favorite players by bringing letter cards so you can spell out their number. Each fan has different favorites, so they each need to bring different sets of letters.

The English spellings for the numbers 0 to 12 are:

ZERO ONE TWO THREE FOUR FIVE SIX
SEVEN EIGHT NINE TEN ELEVEN TWELVE

Input Format:

Read a set of integers from 0 to 12, separated by spaces, representing one fan's favorite players. The last integer will be 999, marking the end of the line.

Output Format:

Print the same numbers, then a period and a space. Then, in alphabetical order, print all the letters the fan needs to be able to spell any one of the jersey numbers provided

▼ Logical Test Cases

Test Case 1 INPUT (STDIN) 2 3 0999	Test Case 2 INPUT (STDIN) 8 14 0999
---	--

EXPECTED OUTPUT

2 3 0999. E E H O R T T W

EXPECTED OUTPUT

8 14 0999. E G H I T

▼ Mandatory Test Cases

Test Case 1

KEYWORD

char nums[13][256]

Test Case 2

KEYWORD

for(n=0;n<26;n++)

▼ Complexity Test Cases

Test Case 1

CYCLOMATIC COMPLEXITY

8

Test Case 2

TOKEN COUNT

270

Test Case 3

NLOC

40

```
1 #include <stdio.h>
2 #include <string.h>
3
4 int main() {
5     char nums[13][256] = {
6         "ZERO", "ONE", "TWO", "THREE", "FOUR", "FIVE", "SIX",
7         "SEVEN", "EIGHT", "NINE", "TEN", "ELEVEN", "TWELVE"
8     };
9     int val;
10    int inputs[100];
11    int count = 0;
12
13    while (scanf("%d", &val) != EOF) {
14        inputs[count++] = val;
15        if (val == 999) {
16            int counts[26] = {0};
17            for (int i = 0; i < count - 1; i++) {
18                int num = inputs[i];
19                if (num >= 0 && num <= 12) {
20                    for (int j = 0; nums[num][j] != '\0'; j++) {
21                        counts[nums[num][j] - 'A']++;
22                    }
23                }
24            }
25        }
26    }
```

```
26         if (inputs[i] == 999) {
27             printf("0999. ");
28         } else {
29             printf("%d ", inputs[i]);
30         }
31     }
32     int first = 1;
33     int n;
34     for (n = 0; n < 26; n++) {
35         while (counts[n] > 0) {
36             if (!first) printf(" ");
37             printf("%c", 'A' + n);
```

```
38         first = 0;
39         counts[n]--;
40     }
41 }
42 printf("\n");
43 count = 0;
44 }
45 }
46 return 0;
47}
48
```

```
C:\Users\CH.SC.U4CSE25225\Desktop\DSA Lab>type t1.txt
```

```
2 3 0999
```

```
C:\Users\CH.SC.U4CSE25225\Desktop\DSA Lab>Array5.exe
```

```
2 3 0999. E E H O R T T W
```

```
C:\Users\CH.SC.U4CSE25225\Desktop\DSA Lab>
```

```
C:\Users\CH.SC.U4CSE25225\Desktop\DSA Lab>type t2.txt
```

```
8 14 0999
```

```
C:\Users\CH.SC.U4CSE25225\Desktop\DSA Lab>Array5.exe
```

```
8 14 0999. E G H I T
```

```
C:\Users\CH.SC.U4CSE25225\Desktop\DSA Lab>
```

Problem Description:

Good news! Suresh get to go to America on a class trip! Bad news, he don't know how to use the Dollar which is the name of the American cash system. America uses coins for cash a lot more than the Kuwait does. Dollar comes in coins for values of: 1, 2, 10, 50, 100, & 500 To practice your Dollar skills, suresh have selected random items from Amazon.co.us and put them into a list along with their prices in Dollar. Suresh now want to create a program to check suresh Dollar math.

Suresh goal is to maximize your buying power to buy AS MANY items as you can with your available Dollar.

Input Format:

File listing 2 to 6 items in the format of:

ITEM DDDDD

ITEM = the name of the item you want to buy

DDDDD = the price of the item (in Dollar)

Output Format:

Print the output in a separate lines contains, List the items suresh can afford to buy. Each item on its own line. Suresh goal is to buy as many items as possible. If suresh can only afford the one expensive item, or 2 less expensive items on a list, but not all three, then list the less expensive items as affordable. If suresh cannot afford anything in the list, output "I need more Dollar!" after the items. The final line you output should be the remaining Dollar he will have left over after make purchases.

Logical Test Cases

Test Case 1

INPUT (STDIN)

6000 3
Phone-case 1486
Candybar 863
Sunglasses 5529

EXPECTED OUTPUT

I can afford Phone-case
I can afford Candybar
I can't afford Sunglasses
3651

Test Case 2

INPUT (STDIN)

2100 3
Camera 69555
TV 76439
iPhone 90000

EXPECTED OUTPUT

I can't afford Camera
I can't afford TV
I can't afford iPhone
I need more Dollar!

Mandatory Test Cases

Test Case 1

KEYWORD

char name[MAX] [LEN];

Test Case 2

KEYWORD

int price[MAX]

Test Case 3

KEYWORD

afford [MAX]

Test Case 4

KEYWORD

for(i=0;i<items;i++)

▼ Complexity Test Cases

Test Case 1

CYCLOMATIC COMPLEXITY

11

Test Case 2

TOKEN COUNT

250

Test Case 3

NLOC

33

```
1 #include <stdio.h>
2 #include <string.h>
3
4 #define MAX 100
5 #define LEN 100
6
7 int main() {
8     int total_dollar, items;
9     if (scanf("%d %d", &total_dollar, &items) != 2) return 0;
10
11     char name[MAX][LEN];
12     int price[MAX];
13     int afford[MAX] = {0};
14     int orig_order[MAX];
15     int sorted_indices[MAX];
16
17     int i, j;
18     for (i = 0; i < items; i++) {
19         scanf("%s %d", name[i], &price[i]);
20         orig_order[i] = i;
21         sorted_indices[i] = i;
22     }
23
24     for (i = 0; i < items; i++) {
25         for (j = i + 1; j < items; j++) {
```

```
26         if (price[sorted_indices[i]] > price[sorted_indices[j]]) {
27             int temp = sorted_indices[i];
28             sorted_indices[i] = sorted_indices[j];
29             sorted_indices[j] = temp;
30         }
```

main | CH.SC.U4CSE25225 | Array6.c

Ln 60, Col 1 Spaces: 4 UTF-8 CRLF C

```
32     }
33
34     int remaining = total_dollar;
35     int bought = 0;
36     for (i = 0; i < items; i++) {
37         int idx = sorted_indices[i];
```

```
38     if (price[idx] <= remaining) {
39         afford[idx] = 1;
40         remaining -= price[idx];
41         bought++;
42     }
43 }
44
45 for (i = 0; i < items; i++) {
46     if (afford[i]) {
47         printf("I can afford %s\n", name[i]);
48     } else {
49         printf("I can't afford %s\n", name[i]);
```

```
50     }
51 }
52
53 if (bought == 0) {
54     printf("I need more Dollar!\n");
55 }
56 printf("%d\n", remaining);
57
58 return 0;
59 }
60
```

C:\Users\CH.SC.U4CSE25225\Desktop\DSA Lab>type t1.txt

6000 3

Phone-case 1486

Candybar 863

Sunglasses 5529

C:\Users\CH.SC.U4CSE25225\Desktop\DSA Lab>Array6.exe

I can afford Phone-case

I can afford Candybar

I can't afford Sunglasses

3651

C:\Users\CH.SC.U4CSE25225\Desktop\DSA Lab>

C:\Users\CH.SC.U4CSE25225\Desktop\DSA Lab>type t2.txt

2100 3

Camera 69555

TV 76439

iPhone 90000

C:\Users\CH.SC.U4CSE25225\Desktop\DSA Lab>Array6.exe

I can't afford Camera

I can't afford TV

I can't afford iPhone

I need more Dollar!

2100

C:\Users\CH.SC.U4CSE25225\Desktop\DSA Lab>

Problem Description:

Nirobi have given a matrix C of size N x M to Rio.

Also Rio are given position of submatrix as X1, Y1 and X2, Y2 inside the matrix.

Now Rio needs to find the sum of all elements inside that submatrix.

Can you help Rio in completing the task assigned by Nirobi.

Constraints:

$$1 \leq T \leq 15$$

$$1 \leq N, M \leq 103$$

$$1 \leq C[N][M] \leq 106$$

$$1 \leq X1, Y1, X2, Y2 \leq M$$

Input Format:

The first line of input contains an integer T denoting the number of test cases.

The first line of each test case is n and m, n is the number of rows and m is the number of columns.

The second line of each test case contains C[N][M].

The third line contains four value of X1, Y1, X2, Y2. X1, Y1 is the top left cell and X2, Y2 is the bottom right cell.

Output Format:

Print the sum of all elements inside that submatrix.

Logical Test Cases

Test Case 1

INPUT (STDIN)

1
3 3
91 8 71 14 2 11 6 51 3
1 2 1 3

EXPECTED OUTPUT

79

Test Case 2

INPUT (STDIN)

1
3 3
9 18 7 14 2 11 6 51 3
1 2 1 3

EXPECTED OUTPUT

25

Mandatory Test Cases

Test Case 1

KEYWORD

int C[m][n];

Test Case 2

KEYWORD

C[i][j]

Complexity Test Cases

Test Case 1

Test Case 2

Test Case 3

CYCLOMATIC COMPLEXITY

6

TOKEN COUNT

195

NLOC

33

```
1 #include <stdio.h>
2
3 int main() {
4     int T;
5     if (scanf("%d", &T) != 1) return 0;
6     while (T--) {
7         int n, m;
8         if (scanf("%d %d", &n, &m) != 2) break;
9         int C[n][m];
10        for (int i = 0; i < n; i++) {
11            for (int j = 0; j < m; j++) {
12                scanf("%d", &C[i][j]);
13            }
14        }
15
16        scanf("%d %d %d %d", &x1, &y1, &x2, &y2);
17
18        long long sum = 0;
19        for (int i = x1 - 1; i <= x2 - 1; i++) {
20            for (int j = y1 - 1; j <= y2 - 1; j++) {
21                sum += C[i][j];
22            }
23        }
24        printf("%lld\n", sum);
25    }
```

main | CH.SC.U4CSE25225 | Array7.c

Ln 28, Col 1 Spaces: 4 UTF-8 CRLF C

```
26     return 0;
```

```
27 }
```


C:\Users\CH.SC.U4CSE25225\Desktop\DSA Lab>type t1.txt

1

3 3

91 8 71 14 2 11 6 51 3

1 2 1 3

C:\Users\CH.SC.U4CSE25225\Desktop\DSA Lab>Array7.exe

79

C:\Users\CH.SC.U4CSE25225\Desktop\DSA Lab>

C:\Users\CH.SC.U4CSE25225\Desktop\DSA Lab>type t2.txt

1

3 3

9 18 7 14 2 11 6 51 3

1 2 1 3

C:\Users\CH.SC.U4CSE25225\Desktop\DSA Lab>Array7.exe

25

C:\Users\CH.SC.U4CSE25225\Desktop\DSA Lab>

DS	Session	Arrays	Question Information	Level 1	Challenge 27
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Problem Description:

Nirobi have given a matrix C of size N x M to Rio.

Also Rio are given position of submatrix as X1, Y1 and X2, Y2 inside the matrix.

Now Rio needs to find the sum of all elements inside that submatrix.

Can you help Rio in completing the task assigned by Nirobi.

Constraints:

$$1 \leq T \leq 15$$

$$1 \leq N, M \leq 103$$

$$1 \leq C[N][M] \leq 106$$

$$1 \leq X1, Y1, X2, Y2 \leq M$$

Input Format:

The first line of input contains an integer T denoting the number of test cases.

The first line of each test case is n and m,n is the number of rows and m is the number of columns.

The second line of each test case contains C[N][M].

The third line contains four value of X1, Y1, X2, Y2. X1, Y1 is the top left cell and X2, Y2 is the bottom right cell.

Output Format:

Print the sum of all elements inside that submatrix.

Logical Test Cases

Test Case 1

INPUT (STDIN)

1
3 3
91 8 71 14 2 11 6 51 3
1 2 1 3

EXPECTED OUTPUT

79

Test Case 2

INPUT (STDIN)

1
3 3
9 18 7 14 2 11 6 51 3
1 2 1 3

EXPECTED OUTPUT

25

Mandatory Test Cases

Test Case 1

KEYWORD

int C[m][n];

Test Case 2

KEYWORD

C[i][j]

Complexity Test Cases

Test Case 1

CYCLOMATIC COMPLEXITY

6

Test Case 2

TOKEN COUNT

195

Test Case 3

NLOC

33

```
1 #include <stdio.h>
2
3 int main() {
4     int T;
5     if (scanf("%d", &T) != 1) return 0;
6     while (T--) {
7         int n, m;
8         if (scanf("%d %d", &n, &m) != 2) break;
9         int C[n][m];
10        for (int i = 0; i < n; i++) {
11            for (int j = 0; j < m; j++) {
12                scanf("%d", &C[i][j]);
13            }
14        }
15
16        scanf("%d %d %d %d", &x1, &y1, &x2, &y2);
17
18        long long sum = 0;
19        for (int i = x1 - 1; i <= x2 - 1; i++) {
20            for (int j = y1 - 1; j <= y2 - 1; j++) {
21                sum += C[i][j];
22            }
23        }
24        printf("%lld\n", sum);
25    }
```

```
26     return 0;
```

```
27 }
```



```
C:\Users\CH.SC.U4CSE25225\Desktop\DSA Lab>type t1.txt
```

```
1
```

```
3 3
```

```
91 8 71 14 2 11 6 51 3
```

```
1 2 1 3
```

```
C:\Users\CH.SC.U4CSE25225\Desktop\DSA Lab>Array8.exe
```

```
79
```

```
C:\Users\CH.SC.U4CSE25225\Desktop\DSA Lab>
```

```
C:\Users\CH.SC.U4CSE25225\Desktop\DSA Lab>type t2.txt
```

```
1
```

```
3 3
```

```
9 18 7 14 2 11 6 51 3
```

```
1 2 1 3
```

```
C:\Users\CH.SC.U4CSE25225\Desktop\DSA Lab>Array8.exe
```

```
25
```

```
C:\Users\CH.SC.U4CSE25225\Desktop\DSA Lab>
```

DS	Session	Arrays	Question Information	Level 1	Challenge 29
Problem Description: Tina has given a boolean matrix mat[P][Q] of size P X Q to Laaysa, She wants to modify it such that if a matrix cell mat[m][n] is 1 (or true) then make all the cells of m_th row and nth column as 1. Can you help Laaysa? Constraints: 1 <= p, q <= 1000 Input Format: First line of the input is the how many rows and columns in Laaysa matrix. After that, each line will represent each row and Laaysa need to enter numbers. Output Format: Print the resultant matrix output in a separate lines.					

Logical Test Cases

Test Case 1

INPUT (STDIN)

2 2
1 0
0 1

EXPECTED OUTPUT

1 1
1 1

Test Case 2

INPUT (STDIN)

3 3
0 0 0
0 1 0
0 0 1

EXPECTED OUTPUT

0 1 1
1 1 1
1 1 1

Mandatory Test Cases

Test Case 1

KEYWORD

int arr[r][c];

Test Case 2

KEYWORD

arrTemp[r][c]

Test Case 3

KEYWORD

for(m=0;m<r;m++)

▼ Complexity Test Cases

Test Case 1

CYCLOMATIC COMPLEXITY

13

Test Case 2

TOKEN COUNT

325

Test Case 3

NLOC

51

```
1 #include <stdio.h>
2
3 int main() {
4     int r, c;
5     if (scanf("%d %d", &r, &c) != 2) return 0;
6
7     int arr[r][c];
8     int arrTemp[r][c];
9     int row_flag[r];
10    int col_flag[c];
11
12    int m, n;
13    for (m = 0; m < r; m++) row_flag[m] = 0;
14    for (n = 0; n < c; n++) col_flag[n] = 0;
15
16    for (m = 0; m < r; m++) {
17        for (n = 0; n < c; n++) {
18            scanf("%d", &arr[m][n]);
19            if (arr[m][n] == 1) {
20                row_flag[m] = 1;
21                col_flag[n] = 1;
22            }
23        }
24    }
25
```

```
26     for (m = 0; m < r; m++) {
27         for (n = 0; n < c; n++) {
28             if (row_flag[m] || col_flag[n]) {
29                 arrTemp[m][n] = 1;
30             } else {
31                 arrTemp[m][n] = 0;
32             }
33         }
34     }
35
```

```
36     for (m = 0; m < r; m++) {
37         for (n = 0; n < c; n++) {
38             printf("%d ", arrTemp[m][n]);
39         }
40         printf("\n");
41     }
42     return 0;
43 }
44
```

C:\Users\CH.SC.U4CSE25225\Desktop\DSA Lab>type t1.txt

2 2

1 0

0 1

C:\Users\CH.SC.U4CSE25225\Desktop\DSA Lab>Array9.exe

1 1

1 1

C:\Users\CH.SC.U4CSE25225\Desktop\DSA Lab>

C:\Users\CH.SC.U4CSE25225\Desktop\DSA Lab>type t2.txt

3 3

0 0 0

0 1 0

0 0 1

C:\Users\CH.SC.U4CSE25225\Desktop\DSA Lab>Array9.exe

0 1 1

1 1 1

1 1 1

C:\Users\CH.SC.U4CSE25225\Desktop\DSA Lab>

Problem Description:

Bear Grylls is a forest lover, so he spends some free time taking care of many of her loved ones' animals. He likes to offer them treats, but wants to do that in an impartial way.

Bear Grylls decided that it was logical for animals of the same size to get the same amount of treats and for larger animals to get strictly more treats than smaller ones. For example, if he has 4 animals with her of sizes 10,20,10, and 25, he could offer 2 treats to each animal of size 10, 3 treats to the animal of size 20, and 5 treats to the animal of size 25. This requires her to buy a total of $2+3+2+5=12$ treats. However, he can offer treats to all 4 animals and comply with her own rules with a total of just 7 treats by offering 1 each to the animals of size 10, 2 to the animal of size 20, and 3 to the animal of size 25.

Help Bear Grylls plan her next animal day. Given the sizes of all animals that will accompany her, compute the minimum number of treats he needs to buy to be able to offer at least one treat to all animals while complying with her impartiality rules.

Constraints:

$$1 \leq T \leq 100.$$

$$1 \leq S_i \leq 100, \text{ for all } i.$$

$$2 \leq N \leq 100.$$

Input Format:

The first line of the input gives the number of test cases, T . T test cases follow.

Each test case consists of two lines.

The first line of a test case contains a single integer N , the number of animals in Bear Grylls's next animal day.

The second line of a test case contains N integers $S_1, S_2, \dots, S_N$, representing the sizes of each animal.

Output Format:

Print the output in a separate lines contains, the minimum number of treats he needs to buy to be able to offer at least one treat to all animals while complying with her impartiality rules.

Logical Test Cases

Test Case 1

INPUT (STDIN)

3
4
10 20 10 25
5
7 7 7 7 7
2
100 1

EXPECTED OUTPUT

7
5
3

Test Case 2

INPUT (STDIN)

5
5
8 6 4 8 9
3
8 5 2
4
101 120 110 215
5
1711 171 711 71 71
2
110 11

EXPECTED OUTPUT

13
6
10
11
3

Mandatory Test Cases

Test Case 1

KEYWORD

int s[MAXN];

Test Case 2

KEYWORD

void sol()

Test Case 3

KEYWORD

read(s[i])

✓ Complexity Test Cases

Test Case 1

CYCLOMATIC COMPLEXITY

13

Test Case 2

TOKEN COUNT

325

Test Case 3

NLOC

51

```
1 #include <stdio.h>
2 #include <stdlib.h>
3
4 #define MAXN 1005
5
6 int s[MAXN];
7 int temp[MAXN];
8
9 int cmp(const void *a, const void *b) {
10     return (*(int *)a - *(int *)b);
11 }
12
13 void sol() {
14     int N;
15     if (scanf("%d", &N) != 1) return;
16     for (int i = 0; i < N; i++) {
17         scanf("%d", &s[i]);
18         temp[i] = s[i];
19     }
20     qsort(temp, N, sizeof(int), cmp);
21
22     int unique[MAXN];
23     int u_count = 0;
24     for (int i = 0; i < N; i++) {
25         if (i == 0 || temp[i] != temp[i - 1]) {
```

```
26         unique[u_count++] = temp[i];
27     }
28 }
29
```

main | CH.SC.U4CSE25225 | Array10.c

Ln 58, Col 1 Spaces: 4 UTF-8 CRLF C

```
31     for (int i = 0; i < N; i++) {
32
33         int low = 0, high = u_count - 1, rank = 1;
34         while (low <= high) {
35             int mid = low + (high - low) / 2;
36             if (unique[mid] == s[i]) {
```

```
37         rank = mid + 1;
38         break;
39     } else if (unique[mid] < s[i]) {
40         low = mid + 1;
41     } else {
42         high = mid - 1;
43     }
44 }
45 total_treats += rank;
46 }
47 printf("%lld\n", total_treats);
```

```
48}  
49  
50int main() {  
51    int T;  
52    if (scanf("%d", &T) != 1) return 0;  
53    while (T--) {  
54        sol();  
55    }  
56    return 0;  
57}  
58
```

```
C:\Users\CH.SC.U4CSE25225\Desktop\DSA Lab>type t1.txt
3
4
10 20 10 25
5
7 7 7 7 7
2
100 1
C:\Users\CH.SC.U4CSE25225\Desktop\DSA Lab>Array10.exe
7
5
3
C:\Users\CH.SC.U4CSE25225\Desktop\DSA Lab>
C:\Users\CH.SC.U4CSE25225\Desktop\DSA Lab>type t2.txt
5
5
8 6 4 8 9
3
8 5 2
4
101 120 110 215
5
1711 171 711 71 71
2
110 11
C:\Users\CH.SC.U4CSE25225\Desktop\DSA Lab>Array10.exe
13
6
10
```



```
C:\Users\CH.SC.U4CSE25225\Desktop\DSA Lab>
```